

Roller Coaster Designer (rollercoaster3)

Eduardo is designing roller coasters. A roller coaster track consists of N sections, numbered from 0 to $N - 1$. Each section is of one of four types (represented by a character):

- ‘D’ – Downward section: increases speed by 9 km/h.
- ‘U’ – Upward section: decreases speed by 11 km/h.
- ‘H’ – Horizontal section: does not change speed.
- ‘B’ – Boost section: increases speed by 10 km/h.

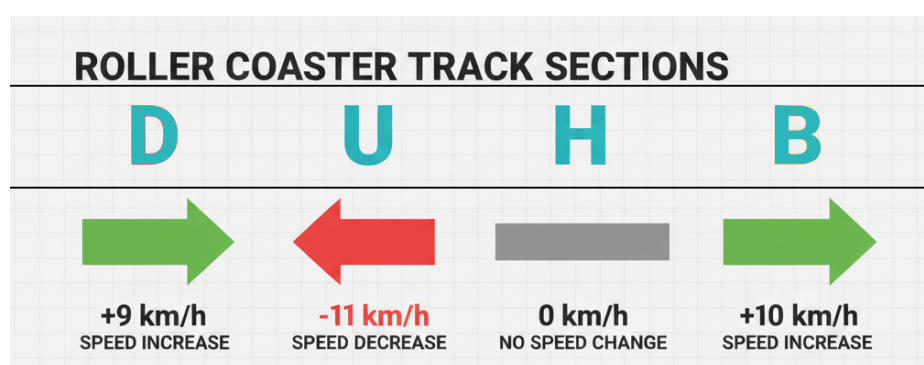


Figure 1: Roller Coaster rules.

A carriage moves along a track, occupying one section at a time. It starts in section 0 with an initial speed of 0 km/h.

A track design is called *plausible* if, after passing section 0, the carriage’s speed remains strictly positive at every section – including after the last one. This means that the first section must be either D or B; otherwise, the speed would become zero or negative.

Eduardo’s current track design is represented by a string S of length N , where each character denotes a section type. He may replace some (possibly zero) **horizontal** (H) sections with **boost** (B) sections to make the design plausible.

Determine the **minimum** number of replacements required to make the design plausible, or report that it is impossible.

 Among the attachments of this task you may find a template file `rollercoaster.*` with a sample incomplete implementation.

Input

The first line contains an integer N , the length of the track.

The second line contains a string S of length N , representing the current roller coaster design.

Output

Print the minimum number of H $\rightarrow$ B replacements to make the design plausible, or -1 if it is impossible.

Constraints

- $1 \leq N \leq 100$.
- S is a string of length N over the alphabet D, U, H, B.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (10 points) There are no upward sections. The first section is a boost section.

- **Subtask 3** (15 points) There are no upward sections.

- **Subtask 4** (20 points) There are no horizontal sections.

- **Subtask 5** (20 points) There are at most 2 horizontal sections.

- **Subtask 6** (35 points) No additional limitations.


Examples

input	output
4 BHUD	1
4 DUHB	-1
5 HHDUU	2

Explanation

In the **first sample case**, section 0 increases the speed to 10 km/h. For section 1, we have two options:

- We do not replace it, so the speed remains 10 km/h. In this case, section 2 decreases the speed to -1 km/h, and the carriage does not reach section 3.
- We replace it with a boost section, so the speed increases to 20 km/h. Then section 2 decreases the speed to 9 km/h, and section 3 increases it to 18 km/h.

In the **second sample case**, the section 0 increases the speed to 9 km/h, and section 1 decreases it to -2 km/h. Thus, it is impossible to make the design plausible.

In the **third sample case**, if we replace both H sections with B sections, the carriage can reach the end. It can be shown that replacing fewer sections is not sufficient.